ECOFEED - INTELLIGENT FOOD DONATION AND WASTE RECYCLING SYSTEM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**ECOFEED-INTELLIGENT FOOD DONATION AND WASTE RECYCLING SYSTEM**" has been carried out by the students, **NANTHINI S, NAVEENA R, PADMA HASSANI N P, PRADEEPA B,** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

EcoFeed is an intelligent mobile application that reduces food waste and hunger by connecting food donors with nearby NGOs and biogas partners. Using GPS-based matching and expiry-date prioritization, edible food is directed to NGOs, while expired food and vegetable waste are routed to biogas and composting partners. Developed using Kotlin and Jetpack Compose, the app uses Supabase for authentication, database management, image storage, and real-time updates, enabling efficient food redistribution and organic waste recycling. The application provides separate user roles for donors, NGOs, and biogas partners to support efficient coordination. It helps donors list surplus food and organic waste with details such as quantity, expiry date, and availability. NGOs can identify suitable food donations and coordinate their collection before the food becomes unsuitable for consumption. Real-time notifications help users receive timely updates about donation requests and collection activities. Overall, EcoFeed aims to reduce unnecessary food disposal, support people facing hunger, and encourage sustainable waste recycling practices.

Keywords: EcoFeed, Food Donation, GPS-Based Matching, Expiry-Date Prioritization, Organic Waste Recycling, Biogas Production, NGOs, MongoDB Atlas, Kotlin, Node.js, Renewable Energy.

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
APK	-	Android Package Kit
NGO	-	Non-Governmental Organization
API	-	Application Programming Interface
GUI	-	Graphical User Interface
SQL	-	Structured Query Language
CPU	-	Central Processing Unit
GPS	-	Global Positioning System
UI	-	User Interface
UX	-	User Experience

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

Every day, large quantities of edible food are discarded by households, restaurants, caterers, hotels and events while many people go without a proper meal. Food that is still safe to eat is often thrown away only because the donor does not know whom to contact, or because there is no quick and reliable way to hand it over before it spoils. At the same time, food that has already expired is usually sent to landfill instead of being converted into something useful such as biogas.

A major problem in the existing situation is the lack of coordination between the three parties involved: the people who have surplus food, the NGOs that can feed people in need, and the biogas partners that can process expired food. Donors rely on phone calls, messaging groups and personal contacts, NGOs do not get a timely and trustworthy view of what is available nearby, and biogas partners rarely learn about expired food at all.

Because of this, donations are delayed, requests go unrecorded, and nobody can later verify what was donated, who accepted it and what finally happened to it. The absence of notifications, status tracking and digital history also makes it difficult to measure how much food has been saved or to improve the process over time.

With the wide availability of smartphones and cloud-based backend platforms, it is now possible to build a mobile application that connects these parties in real time. EcoFeed is an Android food-waste management application developed with Kotlin and Jetpack Compose and supported by Supabase for authentication, a PostgreSQL database, image storage and realtime updates. It redistributes usable surplus food to NGOs and routes expired food to biogas partners.

Many existing food-sharing solutions address only one part of the problem. They may list surplus food but do not separate usable food from expired food, do not enforce who is allowed to do what, or do not keep a permanent record of requests and notifications.

1.2 PROBLEM STATEMENT

Food waste is a serious social, economic and environmental problem. A large share of the food produced for human consumption is never eaten, and a considerable part of it is wasted at the household, retail and food-service stages, where surplus is generated in small, unpredictable quantities. Surplus food that could feed people is lost because the donor cannot reach a suitable NGO in time, while expired food that could generate energy ends up in landfill.

Manual coordination through phone calls and messaging groups is slow and depends on personal contacts. There is no single place where donors can publish a donation with its quantity, expiry date and photograph, no reliable way for NGOs to see only food that is still valid, and no way for biogas partners to receive expired food in a structured manner. Requests and acceptances are not recorded, so responsibility and outcomes cannot be traced.

There is therefore a need for an efficient, low-cost and secure platform that connects food donors, NGOs and biogas partners through a single application. The platform should let donors create donations, show non-expired food to NGOs, show expired food to biogas partners, handle requests and their statuses, notify the users involved and keep a history of all activity.

A further requirement is strict role-based control. Only a food donor should be able to create donations, while NGOs and biogas partners should only view the feed meant for them and raise requests. This restriction must be enforced not only in the Android user interface but also in the database through Row Level Security, so that it cannot be bypassed even if the interface shows an option by mistake.

The project also aims to bridge the gap between voluntary food-sharing efforts and modern technology by providing an accessible mobile solution. By using Supabase Authentication, PostgreSQL, Storage and Realtime, EcoFeed ensures that profiles, donations, images, requests, notifications and history persist across logins and devices, and that changes appear on other devices without logging in again.

1.3 FOOD WASTE MANAGEMENT SYSTEM DOMAIN

Food waste management refers to the process of preventing, redistributing, recycling and recovering surplus and wasted food so that its value is retained for as long as possible. In the commonly used food waste hierarchy, preventing waste and redistributing edible food to people come first, followed by recycling and energy recovery, with disposal as the last option. Digital platforms support this hierarchy by connecting those who have surplus food with those who can use it.

The field includes several subdomains such as surplus food donation, food banking and NGO distribution, expiry tracking, organic-waste processing through anaerobic digestion into biogas, and notification and tracking systems. Each area addresses a different stage of what happens to food once it is no longer needed by its original owner. EcoFeed brings these stages together in one mobile application.

1.3.1 Food Donation Data Management

Food donation data management involves storing and maintaining information about each donation such as the donor, title, quantity, expiry date, photograph, status and the organisation that accepted it. In EcoFeed this information is held in the donations table of a Supabase PostgreSQL database, while the related requests, notifications and activity history are stored in their own tables.

This data is essential because it allows every party to see the current state of a donation and allows the system to decide where it belongs. For example, the expiry date determines whether a donation is shown in the NGO feed or the biogas partner feed, and the stored status shows whether it is available, requested, accepted or completed.

Proper handling of data is important for accuracy, accountability and privacy. Images are kept in Supabase Storage and only their paths or URLs are saved in the database, so that profile and donation photographs persist across logins and devices while the database remains compact and fast.

1.3.2 Role-Based Access and Security

Role-based access control grants permissions according to the role of a user rather than the identity alone. EcoFeed defines three roles: food_donor, ngo and biogas_owner, stored in the profiles table and tied to the authenticated user through profiles.id = auth.users.id. The stored role decides which dashboard a user sees and which actions are allowed.

Security is enforced in the database through PostgreSQL Row Level Security policies. A donor can create donations only when the profile role is food_donor, users can read only their own profiles and notifications, and NGOs or biogas partners cannot insert donations even if the Android interface accidentally displays a Create Donation button.

This two-layer approach, with checks in the application and in the database, keeps the system safe and trustworthy. Secret and service-role keys are never placed in the Android application, and invalid or missing roles are rejected rather than silently treated as another role.

1.3.3 Realtime Redistribution Using Supabase

Realtime redistribution means that a change made by one user is visible to the other users within moments, without logging in again. EcoFeed uses Supabase Realtime together with Kotlin Coroutines and StateFlow so that new donations, requests, status changes and notifications appear on other devices as they happen.

This reduces the delay between a donation being posted and an NGO or biogas partner responding to it, which is critical for food that spoils quickly. The expiry workflow automatically sends food with a future expiry date to the NGO feed and food that has already expired to the biogas partner feed.

Persistent notifications, request statuses and activity history make the system practical and transparent. Together they give donors, NGOs and biogas partners a single platform that reduces food waste, supports people in need and turns unusable food into a renewable energy resource

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The aim of this project is to develop EcoFeed, an Android food-waste management application that connects food donors, NGOs and biogas partners, redistributes usable surplus food to NGOs and routes expired food to biogas partners, using Kotlin, Jetpack Compose and Supabase.

The project also aims to enforce secure role-based access so that each user sees the correct dashboard and can perform only the actions permitted for their role, and to ensure that profiles, donations, images, requests, notifications and history are stored persistently and updated in realtime across devices.

1.4.2 Objectives

- To provide registration and login with Supabase Authentication and assign one of three roles: food donor, NGO or biogas partner.
- To allow food donors to create and manage donations with title, quantity, expiry date and photograph.
- To route non-expired food to the NGO feed and expired food to the biogas partner feed.
- To implement request, acceptance, rejection and status workflows for NGOs and biogas partners.
- To store notifications and activity history persistently and deliver updates in real time across devices.
- To secure the data with PostgreSQL Row Level Security and block donation creation by NGOs and biogas partners.
- To provide separate dashboards and options based on the user's assigned role, making the application easier to use.
- To maintain accurate donation records so users can track requests and review previous activities.

- To improve coordination between donors, NGOs and biogas partners through a centralized digital platform.
- To encourage responsible food distribution and the effective utilization of organic waste.

1.5 SCOPE OF THE PROJECT

The scope of this project covers the development of an Android application, built with Kotlin and Jetpack Compose, that serves three types of users: food donors, NGOs and biogas partners. The application uses Supabase as its backend for authentication, database, image storage and real-time updates, and keeps the existing user interface design.

Donors can register, create donations with photographs and track their donation history and incoming requests. NGOs can view valid, non-expired donations, request food and receive notifications. Biogas partners can view and process expired donations, receive notifications and track processing history. The application also maintains donation statuses to help users understand whether a donation is available, requested, accepted, rejected or completed.

The system can be used in:

- Restaurants, hotels, caterers and event organisers with surplus food
- NGOs, food banks and community kitchens
- Biogas plants and organic-waste processing partners
- Educational and research projects on food waste and sustainable technology

The proposed solution reduces food waste, speeds up redistribution, supports renewable energy recovery and provides a transparent record of every donation and request. It also helps improve communication between different user groups and reduces dependence on manual coordination. The modular design allows additional features to be introduced in the future without changing the entire application.

CHAPTER 2

LITERATURE SURVEY

2.1 SMART FOOD DONATION AND WASTE REDUCTION SYSTEM THANGADURAI M, HARESHWAR M S, MANONMANI R, MANIMARAN R (2026)

Food waste is a major global concern, as large quantities of edible food are discarded daily while many people continue to suffer from hunger. Traditional food donation systems depend on manual coordination and delayed communication, resulting in inefficient food distribution and limited outreach. The proposed Smart Food Donation and Waste Reduction System is a web-based application designed to connect restaurants, supermarkets, and non-governmental organizations (NGOs) for the effective redistribution of surplus food.

The paper provides separate interfaces for food donors and receivers, supported by a centralized cloud database for efficient data management. It integrates the Google Maps API to identify nearby NGOs based on real-time location matching, helping reduce transportation time and improve food collection efficiency. Instant notifications and automated alerts enable receivers to collect available food before it spoils.

The application processes information such as food type, quantity, and pickup time through RESTful APIs. It also includes analytics dashboards to monitor donation trends and measure food waste reduction. By improving communication between donors and receivers, the system supports faster food redistribution and reduces the amount of edible food discarded.

Overall, the system offers a technology-driven approach to food donation by improving coordination, optimizing collection, and supporting more efficient resource utilization. It contributes to the United Nations Sustainable Development Goal of Zero Hunger by helping surplus food reach people in need.

2.2 BIOTRANSFORMATION OF FOOD WASTE INTO BIOGAS VIA MICROBIAL ANAEROBIC DIGESTION RAJEEV SINGH, DAN BAHADUR PAL, BASANT LAL, TRIPTI SINGH, IRFAN AHAMD, ABDULLAH MASHRAQI, SHAFIUL HAQUE, NEHA SRIVASTAVA (2025)

The Smart Food Waste Management System is a platform designed to reduce food wastage by connecting food donors, NGOs, and delivery personnel through web and mobile applications. It enables donors to share surplus food and helps ensure that it reaches people in need efficiently.

The system provides real-time donation tracking, NGO coordination, pickup and delivery management, and an integrated chatbot for instant user support. A centralized database securely maintains user details, donation records, delivery information, and feedback.

The platform promotes sustainability and social responsibility by reducing food waste, improving food redistribution, and minimizing environmental impact. It supports the United Nations Sustainable Development Goals, particularly Zero Hunger (SDG 2) and Responsible Consumption and Production (SDG 12).

Food waste is a major global issue that contributes to environmental pollution and increases food insecurity. The paper proposes a smart food waste management system that connects food donors, non-governmental organizations (NGOS), and delivery personnel through a web-based and mobile application. The main objective is to reduce food wastage by ensuring that surplus food reaches people in need efficiently.

The paper focuses on improving food donation coordination through user-friendly interfaces and efficient data management. By simplifying the donation and delivery process, it aims to minimize food wastage, support people facing hunger, and encourage environmental responsibility. The paper highlights how digital technology can help create a more organized and effective food redistribution process, contributing to sustainability and community well-being.

2.3 THE SMART FOOD WASTE MANAGEMENT SYSTEM IRFAN MOHIUDDIN, MOHAMMED YOUSUF SHAREEF, FARHAN ARIF, MS. RAHMA BAMASDOOS (2025)

Food waste is a major global issue that contributes to environmental pollution and increases food insecurity. The paper proposes a smart food waste management system that connects food donors, non-governmental organizations (NGOS), and delivery personnel through a web-based and mobile application. The main objective is to reduce food wastage by ensuring that surplus food reaches people in need efficiently.

The proposed system enables donors to share details about available surplus food, while NGOs coordinate collection and redistribution. Real-time donation tracking and delivery task management help improve communication between donors and recipients. A chatbot is also integrated to provide instant assistance and answers to frequently asked questions. A centralized database maintains user information, donation records, and feedback to support organized operations and improve the overall user experience

The system focuses on improving food donation coordination through user-friendly interfaces and efficient data management. By simplifying the donation and delivery process, it aims to minimize food wastage, support people facing hunger, and encourage environmental responsibility. The paper highlights how digital technology can help create a more organized and effective food redistribution process, contributing to sustainability and community well-being

2.4 REVIEW IN ANAEROBIC DIGESTION OF FOOD WASTE KEFANG HE, YING LIU, LONGJIN TIAN, WANYOU HE, QUNPENG CHENG (2024)

Food waste is generated from households, schools, restaurants, canteens, and food-processing industries. Improper disposal of food waste can cause environmental pollution, unpleasant odours, and increased pressure on waste-management systems. Anaerobic digestion is an effective method of treating organic waste by breaking down biodegradable materials in the absence of oxygen. This process produces biogas, which can be used to generate electricity and heat, while the remaining digestate can be processed for use as organic fertilizer. The study reviews how food waste can be converted into useful resources and highlights the importance of improving its treatment to support sustainable waste management.

The authors explain several challenges affecting anaerobic digestion, including foaming, acidification, ammonia inhibition, and the presence of emerging contaminants such as antibiotics and microplastics. These problems can disturb microbial activity, reduce methane production, and make the digestion process unstable. To address these issues, the review discusses three major improvement methods: pretreatment, co-digestion, and the addition of suitable materials. Pretreatment helps break down complex organic substances, while co-digestion combines food waste with other organic materials to improve nutrient balance and process stability. Additives such as biochar and iron-based materials may also support microbial activity, reduce the accumulation of harmful substances, and improve biogas production under suitable conditions.

The paper concludes that anaerobic digestion has considerable potential for reducing food waste and recovering renewable energy, but further research is required to improve efficiency, reduce operating costs, and control environmental risks. The authors highlight the need to optimize treatment methods, identify suitable materials for co-digestion, improve the safe use of digestion by-products, and evaluate the overall environmental and economic benefits.

2.5 TRANSFORMING FOOD WASTE INTO ENERGY A COMPREHENSIVE REVIEW SHAMA KAKKAR, NAGARAJU DHARAVAT, SURESH KUMAR SUDABATTULA (2024)

Food waste management has become an important global challenge because large quantities of discarded food contribute to environmental pollution, landfill accumulation, and the loss of valuable resources. Fruit and vegetable waste, leftover food, and other biodegradable materials contain organic compounds that can be recovered and converted into useful forms of energy. The study reviews different technologies for transforming food waste into energy and explains how these technologies can contribute to sustainable waste management. Instead of considering food waste only as a disposal problem, the paper presents it as a potential resource for renewable energy generation.

The paper discusses different technologies that can be used to recover energy from organic waste, including anaerobic digestion and microbial fuel cells. Anaerobic digestion uses microorganisms to break down biodegradable materials in oxygen-free conditions and produces biogas, which can be used for heat and electricity generation. Microbial fuel cells use microorganisms to convert chemical energy stored in organic matter into electrical energy under suitable operating conditions. The selection of an appropriate technology depends on the availability and characteristics of the feedstock, the desired energy output, processing requirements, and economic feasibility. The study also highlights practical difficulties associated with food waste collection, transportation, segregation, storage, and processing. Variations in waste composition can affect treatment efficiency, while infrastructure requirements, operating costs, and the availability of suitable facilities can influence implementation. Effective waste management policies, appropriate technology selection, and cooperation among waste generators, collection systems, and processing facilities are therefore important for successful energy recovery.

2.6 RECYCLING, RECOVERING AND PREVENTING "FOOD WASTE": COMPETING SOLUTIONS FOR FOOD SYSTEMS SUSTAINABILITY IN THE UNITED STATES AND FRANCE - MARIE MOURAD (2016)

Food waste is a major environmental, social, and economic challenge across the world. This study examines different approaches to food waste management in the United States and France, focusing on how organizations handle surplus food and reduce unnecessary waste. It explores the relationship between food waste prevention, food recovery, and recycling, highlighting the importance of developing sustainable practices throughout the food system. Reducing food waste can also help conserve natural resources and improve the availability of food for people in need.

The paper discusses three major approaches to managing food waste: preventing surplus food at its source, recovering edible food for human consumption, and recycling remaining waste into useful products such as animal feed, energy, and compost. Food recovery can help redirect edible surplus food to charities and people experiencing food insecurity, while recycling provides alternative uses for organic materials that cannot be consumed. The study also explains that different organizations may have different priorities depending on their environmental, social, and economic objectives.

The paper emphasizes that preventing food waste before it occurs is an important step towards long-term sustainability. Better planning, improved distribution systems, responsible purchasing, and effective food recovery practices can help reduce the quantity of food discarded. It also highlights the need for cooperation among businesses, communities, and waste management organizations to improve existing practices. By combining prevention, recovery, and recycling, food systems can reduce environmental impacts, use resources more efficiently, and support a more sustainable future.

2.7 REVIEWING THE ANAEROBIC DIGESTION OF FOOD WASTE FOR BIOGAS PRODUCTION CUNSHENG ZHANG, HAIJIA SU, JAN BAEYENS, TIANWEI TAN(2014)

Food waste is a significant environmental concern because its improper disposal can cause pollution and waste valuable resources. This study reviews anaerobic digestion as an effective method of converting food waste into biogas. It explains the characteristics of food waste and the basic principles of anaerobic digestion, in which microorganisms break down organic materials in the absence of oxygen. The resulting biogas can be used for renewable energy production, helping reduce dependence on conventional fuels.

The paper examines factors that influence the efficiency of anaerobic digestion, including the composition of food waste, process conditions, and the stages involved in breaking down organic materials. It discusses pretreatment methods that can improve the digestion process and help microorganisms access organic matter more effectively. The study also explores co-digestion, where food waste is processed with other organic materials to improve nutrient balance and potentially increase biogas and methane production.

The paper highlights the potential of anaerobic digestion as a sustainable food waste management solution. Converting organic waste into biogas can reduce the environmental problems associated with uncontrolled disposal while recovering useful energy. These findings are relevant to EcoFeed because the application directs suitable organic food waste towards biogas partners rather than treating all surplus food as disposable waste. This approach can support better waste segregation, resource recovery, and environmentally responsible food waste management.

2.8 THE FOOD WASTE HIERARCHY AS A FRAMEWORK FOR THE MANAGEMENT OF FOOD SURPLUS AND FOOD WASTE EFFIE PAPARGYROPOULOU, RODRIGO LOZANO, JULIA K. STEINBERGER, NIGEL WRIGHT, ZAINI BIN UJANG (2014)

Food waste has become a major global concern because of its environmental, social, and economic impacts. The study introduces a food waste hierarchy framework to identify effective ways to prevent and manage surplus food and food waste throughout the food supply chain. It highlights the importance of sustainable production and consumption practices to reduce unnecessary food waste and conserve natural resources.

The proposed framework prioritizes preventing food waste by reducing surplus food and avoidable waste. When surplus food is available, it should be redistributed to people experiencing food insecurity rather than being discarded. The study also discusses alternative uses for food waste, such as converting it into animal feed, to make better use of available resources and reduce the amount of waste sent for disposal.

The framework considers environmental protection, economic benefits, and social well-being when deciding how food surplus and waste should be managed. It emphasizes the need to improve existing practices across the food supply chain and promote responsible resource utilization. By prioritizing food waste prevention, redistribution, and appropriate recovery methods, the framework supports a more sustainable food system and helps reduce the negative impacts of food waste.

This paper is relevant to EcoFeed – Intelligent Food Donation and Waste Recycling System, which follows a similar priority-based approach. EcoFeed aims to direct edible surplus food from donors to NGOs for redistribution, while suitable non-edible food waste and organic kitchen waste can be directed to biogas or composting partners.

2.9 RECYCLING, RECOVERING AND PREVENTING "FOOD WASTE"- COMPETING SOLUTIONS FOR FOOD SYSTEMS SUSTAINABILITY IN THE UNITED STATES AND FRANCE - MARIE MOURAD(2010)

Food waste is a major environmental, social, and economic challenge across the world. This study examines different approaches to food waste management in the United States and France, focusing on how organizations handle surplus food and reduce unnecessary waste. It explores the relationship between food waste prevention, food recovery, and recycling, highlighting the importance of developing sustainable practices throughout the food system. Reducing food waste can also help conserve natural resources and improve the availability of food for people in need.

The paper discusses three major approaches to managing food waste: preventing surplus food at its source, recovering edible food for human consumption, and recycling remaining waste into useful products such as animal feed, energy, and compost. Food recovery can help redirect edible surplus food to charities and people experiencing food insecurity, while recycling provides alternative uses for organic materials that cannot be consumed. The study also explains that different organizations may have different priorities depending on their environmental, social, and economic objectives.

The paper emphasizes that preventing food waste before it occurs is an important step towards long-term sustainability. Better planning, improved distribution systems, responsible purchasing, and effective food recovery practices can help reduce the quantity of food discarded. It also highlights the need for cooperation among businesses, communities, and waste management organizations to improve existing practices. By combining prevention, recovery, and recycling, food systems can reduce environmental impacts, use resources more efficiently, and support a more sustainable future.

2.10 BIOGAS PRODUCTIONCURRENT STATE AND PERSPECTIVES PETERWEILAND(2009)

Biogas production is an important method of converting organic materials into renewable energy while reducing the amount of waste sent for disposal. This study reviews the development of biogas technology and its applications in sustainable energy production. Organic materials such as agricultural residues, animal manure, food waste, and other biodegradable materials can be used as raw materials for producing biogas. The increasing generation of organic waste has created a need for effective treatment methods that can reduce environmental pollution while recovering useful resources. Biogas production provides an alternative to conventional fossil fuels and helps improve the utilization of biological resources. It also supports the principles of a circular economy by converting materials that might otherwise be discarded into valuable energy resources. The technology can be applied in agricultural activities, food-processing industries, restaurants, and other facilities that generate biodegradable waste.

The paper explains anaerobic digestion, a biological process in which microorganisms break down organic matter in the absence of oxygen to produce biogas. The process generally involves four major stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. During these stages, complex organic materials are gradually converted into simpler compounds, which are ultimately used to produce methane and carbon dioxide. The methane content makes biogas useful for generating heat and electricity, depending on the gas quality and available equipment. The efficiency of the process depends on several factors, including the composition of the raw materials, temperature, pH, retention time, organic loading rate, and system design. Proper waste segregation and preparation can also influence the effectiveness of digestion. Maintaining suitable operating conditions and monitoring the digestion process are important for reducing instability, improving gas production, and ensuring the reliable operation of biogas plants.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

At present, surplus food is mostly handled through informal and manual methods. A donor who has leftover food after a function or a day of business usually phones a known NGO, posts in a messaging group or simply discards the food. Whether the food reaches someone who needs it depends largely on personal contacts and on luck.

Some general-purpose messaging and social-media tools are used to announce surplus food, and a few food-sharing applications exist. However, these tools are not designed for the specific workflow of donor, NGO and biogas partner. Posts are mixed with other messages, there is no structured information about quantity and expiry, and there is no way to verify who is responding.

In the current practice, an NGO must be contacted one by one, pick-up is arranged by call, and there is no common record of the transaction. Expired food is not separated from usable food in any systematic way and is generally thrown away together with other waste.

Although these methods work on a small scale, they lack role-based access, expiry-based routing, status tracking, notifications and realtime updates. A donor cannot easily know whether a donation has been accepted, and an NGO cannot easily know whether the food is still safe to collect.

Furthermore, the existing system provides no integrated place to store donations, requests and history, and no secure control over who can add or change information. Therefore, a structured, secure and realtime platform is required.

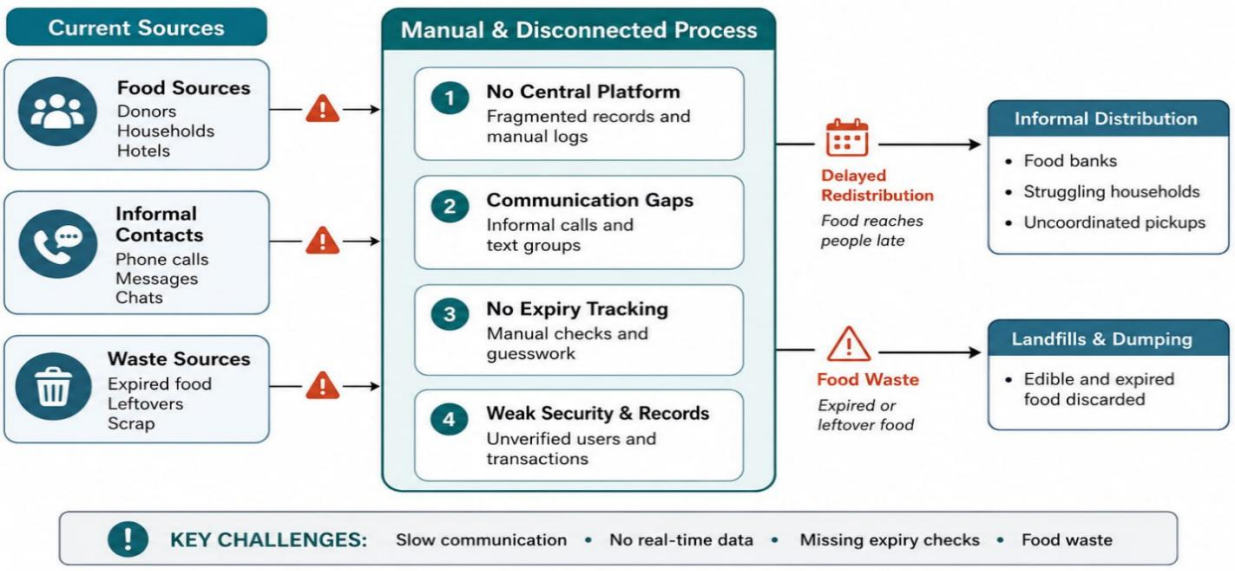


Figure 3.1 Existing System

3.2 DISADVANTAGES

- Depends on phone calls and messaging groups, which are slow and unreliable.
- No single platform connecting food donors, NGOs and biogas partners.
- Donation details such as quantity and expiry date are not recorded in a structured form.
- Usable food and expired food are not separated, so edible food may reach the wrong place.
- Expired food is usually discarded instead of being sent for biogas production.
- No status tracking, so donors do not know whether a donation was accepted.
- No instant notifications to inform NGOs and partners about new donations.
- No digital history of donations, requests and outcomes for accountability.
- No role-based access control, so any user can post or change information.
- No photographs of the food, making it hard to judge its condition before pick-up.

CHAPTER 4

PROBLEMS IDENTIFIED

In the present practice, donors and receivers of surplus food are connected only through informal channels. A donor must find a suitable NGO, call or message them and wait for a reply, and the food may spoil in the meantime. This process is slow, depends on personal contacts and is difficult to repeat reliably.

Where digital tools are used, they are usually general messaging applications or listing platforms that do not distinguish between food that is still valid and food that has expired. As a result, usable food may be sent for disposal, while expired food may be offered to organisations that cannot use it.

Another challenge is the lack of controlled access. When any logged-in user can read or modify everything, donations can be created by the wrong type of user and records can be changed without authorisation. In an earlier version of EcoFeed, NGO and biogas partner accounts sometimes opened the Food Donor dashboard with the Create Donation option, and invalid role values caused the message "Account role is not assigned or invalid".

Data handling was also inconsistent in the earlier versions. Some profiles had invalid roles such as user, several authentication users had missing role metadata, policies on profiles, notifications and requests were duplicated or conflicting, and old MongoDB and Express calls produced ObjectId-casting errors when given Supabase UUIDs. Such inconsistencies make it difficult to persist profiles, images, notifications and history correctly.

Finally, without realtime updates and stored notifications, users must keep refreshing or logging in again to see changes. Missed requests and delayed responses reduce the chance of saving food in time. These problems motivate a single, secure, role-aware application with a consistent Supabase backend.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system, EcoFeed, is an Android application that connects food donors, NGOs and biogas partners through a Supabase backend. Users register and log in with Supabase Authentication, and the role stored in their profile decides which dashboard and permissions they receive. Donors publish surplus food, the system routes it according to its expiry date, and NGOs or biogas partners respond with requests that are tracked until completion.

The application stores donation information, including the title, quantity, expiry date, photograph and current status, in the database. Based on the expiry date and donation details, suitable food is displayed to NGOs, while expired food is made available to biogas partners for appropriate processing.

Supabase PostgreSQL stores application records, Supabase Storage maintains donation images, and Supabase Realtime supports updates across connected devices.. This centralized approach improves record maintenance, supports reliable communication and makes the donation process easier to monitor. The system is designed to provide a simple interface while maintaining clear separation between donor, NGO and biogas partner activities.

5.2 DONATION CREATION AND IMAGE UPLOAD

Donation creation is the first stage of the workflow and is available only to users with the food donor role. The donor enters the title, quantity and expiry date of the food and attaches a photograph taken with the phone camera or chosen from the gallery.

The photograph is uploaded to the donation-images bucket in Supabase Storage, and the resulting URL or path is saved with the donation record in the donations table.

Profile photographs are handled in the same way through the profile-images bucket, so they remain available after logout and on other devices.

Before saving, the application validates the input, for example by checking that required fields are filled and that the expiry date is valid.

5.3 ROLE-BASED ROUTING AND FOOD EXPIRY WORKFLOW

After login, the application reads the role from the profiles table and normalises it to one of three canonical values: food_donor, ngo or biogas_owner. The navigation graph then opens the dashboard that belongs to that role. A missing or invalid role is rejected with an error message and never defaults to the donor or NGO dashboard.

The expiry date of each donation decides where it is shown. Food with a future expiry date appears in the NGO feed, while food that has already expired appears in the biogas partner feed. In this way usable food is offered to people first and expired food is directed to energy recovery.

NGOs and biogas partners cannot create donations. This is blocked in the Android application by not showing the option and in the database by Row Level Security, so that the restriction holds even if the interface were to show the button by mistake.

5.4 REQUESTS, NOTIFICATIONS AND REALTIME UPDATES

An NGO can request a valid donation and a biogas partner can request an expired one. The request is saved in the requests table with its status, and the donor is notified. The donor can accept or reject the request, and the status change is saved and notified back to the requester.

Notifications are stored in the notifications table so that each user sees only their own updates, and important events are stored in the activity_history table. Because the data is stored in Supabase, notifications, statuses and history persist after logout and across devices.

Supabase Realtime pushes changes to subscribed devices, so that a new donation, request or status update appears on another phone without the user logging in again. Row Level Security ensures that each user receives only the rows they are permitted to see.

Together, these features make a complete workflow: donation, routing, request, acceptance or rejection, notification and history. The Android interface remains simple, while the backend handles security, storage and synchronisation.

5.5 PROPOSED SYSTEM ARCHITECTURE

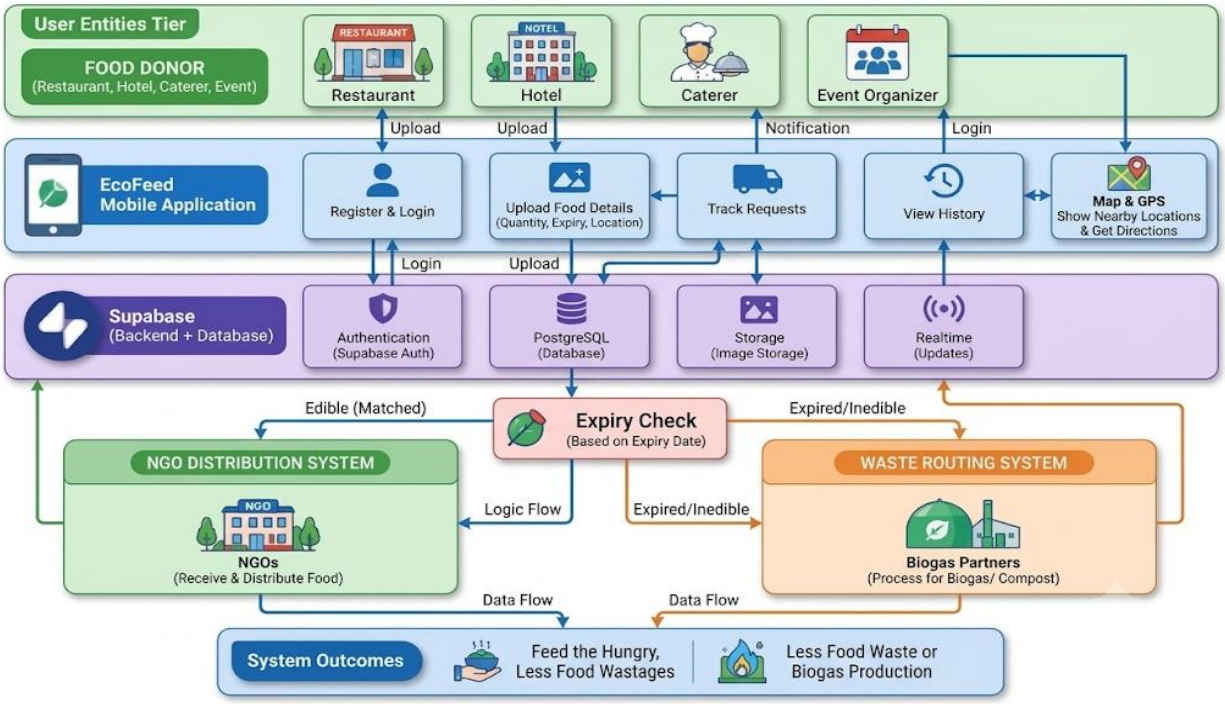


Figure 5.5 Proposed System

The overall workflow begins when a food donor logs in and creates a donation with a photograph. The donation is stored in Supabase and routed by expiry date to the NGO feed or the biogas partner feed. An NGO or biogas partner views the feed that matches their role and sends a request. The donor accepts or rejects it, notific6ations are delivered, and the status and history are updated in realtime on all connected devices.

5.6 ADVANTAGES

- Connects food donors, NGOs and biogas partners on a single platform.
- Sends usable food to NGOs and expired food to biogas partners automatically.
- Enforces role-based access in both the Android app and the database using Row Level Security.
- Persists profiles, images, donations, requests, notifications and history in Supabase.
- Provides realtime updates across devices without requiring users to log in again.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

- Processor
- RAM (Minimum 8 GB)
- SSD / Hard Disk
- Computer / Laptop
- Android Smartphone
- Display Monitor
- Keyboard
- Mouse / Touchpad
- Wi-Fi Router / Mobile Hotspot
- Camera

6.2 SOFTWARE REQUIREMENTS

- Android Studio
- Kotlin Programming Language
- Android SDK
- Jetpack Compose
- Gradle Build System
- Kotlin Coroutines and StateFlow
- Supabase Authentication
- Supabase PostgreSQL Database
- Supabase Storage and Realtime

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The proposed system is implemented using several functional modules. Each module performs a specific task in the overall workflow of EcoFeed. The modular design makes the application easier to develop, test and maintain. The modules are listed below.

- User Authentication and Role Management Module
- Food Donor Module
- Smart Expiry & Automated Waste Routing
- Beneficiary Operations (NGO & Biogas)
- Rating &Feedback System
- Real-Time Notification Engine
- User Profile Management
- UI / UX & Adaptive Theming

7.2 MODULE DESCRIPTION

EcoFeed – Intelligent Food Donation and Waste Recycling System consists of eight integrated modules designed to support efficient food donation, waste segregation, and resource recovery. The User Authentication and Role Management Module manages user registration, login, authentication, and access based on roles such as food donor, NGO, and biogas partner. The Food Donor Module enables donors to create and manage food donation listings by providing details such as food name, quantity, location, availability, expiry information, and photographs. The Smart Expiry and Automated Waste Routing Module prioritizes donations according to expiry information and helps direct edible surplus food towards NGOs while routing suitable non-edible food waste and organic kitchen waste towards biogas or composting partners.

7.2.1 User Authentication and Role Management Module

This module is the entry point of the application. It handles registration and login with Supabase Authentication and creates the matching row in the profiles table, where the profile id equals the authenticated user id. During registration the user selects one of the three roles: food donor, NGO or biogas partner.

After login, the module reads the role from public.profiles.role and normalises it to the canonical values food_donor, ngo or biogas_owner. Navigation then opens the dashboard for that role. An invalid or missing role produces an error and never falls back to another dashboard.

7.2.2 Food Donor Module

The Food Donor module allows restaurants, hotels, caterers, and other food providers to donate their surplus food through the EcoFeed application. Donors can create food listings by entering details such as food name, category, quantity, expiry date, and pickup location. They can also upload food images to provide beneficiaries with a better understanding of the available items. The donation information is stored in the Supabase PostgreSQL database. This makes the listed food available to the appropriate beneficiaries.

The module also enables donors to monitor the progress of their donations from creation to collection. Donors can view whether a listing is available, requested, accepted, or collected, depending on the implemented workflow. They can manage their own listings and review previous donation activities. Status updates help donors understand how their surplus food is being utilized. By simplifying donation management, this module encourages more restaurants and food providers to participate in food redistribution.

7.2.3 Smart Expiry and Automated Waste Routing

The Smart Expiry and Automated Waste Routing module categorizes food based on its expiry date and intended use. The application compares the recorded expiry date with the current system date to determine whether the listing is expired.

Food that has not expired is displayed in the NGO feed for potential redistribution.

Expired food is directed to the Biogas Plant feed for appropriate organic waste processing. This process helps separate food donation activities from waste recycling operations.

The module reduces the need for beneficiaries to manually identify suitable listings from a combined feed. NGOs can focus on available, unexpired food, while biogas partners can identify organic waste suitable for processing. Expiry-based filtering helps prioritize the distribution of food that needs timely collection. However, an expiry date alone cannot establish food safety, so the food's condition and handling requirements must also be checked. This module supports better resource utilization and promotes sustainable food waste management.

7.2.4 Beneficiary Operations

The Beneficiary Operations module provides dedicated functionality for NGOs and biogas plant operators. NGOs can browse available, unexpired food listings and select donations that match their requirements. Biogas partners can view organic waste listings intended for biogas production or other suitable recycling processes. Each listing provides relevant information, including quantity, location, expiry details, and donor contact information where available. This helps beneficiaries identify appropriate resources for collection.

The module enables beneficiaries to submit requests or claim suitable listings according to the application's donation workflow. A one-tap phone calling feature allows beneficiaries to contact donors and coordinate collection arrangements directly. Request and collection status changes are recorded in Supabase so that the relevant users can track progress. Role-specific access ensures that NGOs and biogas partners receive listings appropriate to their activities. This module improves coordination between donors and beneficiaries while supporting food recovery and organic waste recycling.

7.2.5 Ratings & Feedback System

The Ratings and Feedback module allows beneficiaries to share their experiences after a donation pickup has been completed. Users can submit ratings through an interactive five-star review interface.

The feedback may reflect the quality of communication, coordination, and overall collection experience.

Ratings are associated with the relevant donor and completed donation activity. This information helps build transparency among participants in the EcoFeed ecosystem**.**

The module can calculate and display an overall donor rating based on the reviews received. Users can refer to this rating when evaluating the reliability of potential donation partners. Feedback also helps donors understand their performance and identify areas for improvement. To maintain fairness, the application should restrict reviews to eligible users who have completed the corresponding pickup. By encouraging accountability and constructive feedback, this module helps strengthen trust between donors, NGOs, and biogas partners.

7.2.6 Real-Time Notification Engine

The Real-Time Notification Engine keeps users informed about important events occurring within the EcoFeed application. Notifications may be generated when a donation is requested, accepted, collected, or reviewed. Relevant updates are directed to the users involved in the corresponding activity. Supabase PostgreSQL stores notification records, while Supabase Realtime can help the application receive database changes without repeatedly refreshing the screen. This enables users to view updates more conveniently.

The module also maintains notification state to help prevent the same event from producing repeated pop-up banners on a device. Users can review relevant notifications and stay informed about their donation activities. Notifications reduce the need for donors and beneficiaries to repeatedly check listing pages for status changes. Appropriate access controls ensure that users only retrieve notifications intended for them. This module improves communication, supports timely collection, and provides a more responsive application experience.

7.2.7 User Profile Management

The User Profile Management module allows users to maintain their personal and organizational information within EcoFeed. Users can update details such as their name, email information where supported, phone number, organization name, and address. The profile also records the user's assigned application role and other relevant account information. Profile data is stored in the Supabase PostgreSQL database and associated with the authenticated user's unique ID. This helps maintain a consistent profile across sessions and devices.

The module also supports selecting and uploading profile photographs. Images can be compressed on the Android device before upload to reduce file size and improve upload efficiency. Supabase Storage is used to store profile images, while their corresponding paths or URLs are saved in the profile record. Users can view their saved profile information when they log in again.

7.2.8 UI/UX &Adaptive Theming

The UI/UX and Adaptive Theming module focuses on providing a clear, consistent, and user-friendly experience across the EcoFeed Android application. The interface is developed using Kotlin and Jetpack Compose, with Material 3 design components. Screens are organized according to the responsibilities of Food Donors, NGOs, and Biogas Plant operators. Clear navigation, readable text, appropriate spacing, and recognizable controls help users complete tasks with less confusion. The design aims to make food donation and waste collection accessible to different types of users.

The application supports light and dark themes to improve viewing comfort under different lighting conditions. Smooth page transitions help users move between dashboards, food listings, profiles, and activity screens. A custom adaptive launcher icon provides a recognizable identity for EcoFeed on Android devices. Consistent visual components help maintain the same design language throughout the application. This module improves usability and gives users a more organized experience when managing donations, requesting food, or coordinating organic waste collection.

CHAPTER 8
SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is the process of testing individual components of the application separately to ensure that each performs its intended function. In this project, units such as the role normalisation function, the expiry check that decides between the NGO and biogas feeds, the input validation of the donation form and the ViewModels are tested independently.

The purpose of unit testing is to find errors at an early stage. Test cases include valid roles, invalid roles such as user, empty roles, donations expiring in the future and donations that have already expired. An invalid role must be rejected and must not map to any dashboard.

8.2 INTEGRATION TESTING

Integration testing is performed after the individual units work correctly. In this phase, the Android application, Supabase Authentication, the PostgreSQL tables, Storage and Realtime are tested together as a unified system**.**

Integration tests verify, for example, that a newly registered user gets a profile with the correct role, that a donation image uploaded to Storage is saved as a URL in the donation record, that a request creates a notification for the donor, and that Row Level Security rejects a donation insert from an NGO or biogas partner account.

8.3 SYSTEM TESTING

System testing verifies the complete application against its functional and non-functional requirements. All three roles are tested from login to logout, including account switching on the same device.

During system testing, a food donor creates donations with future and past expiry dates, an NGO checks that only valid donations appear, and a biogas partner checks that only expired donations appear. Requests, acceptance, rejection, notifications and history are checked, and the Create Donation option must be absent for NGO and biogas partner accounts.

8.4 PERFORMANCE TESTING

Performance testing evaluates how the application behaves under different loads. It measures the time taken to log in, load a feed, upload an image and receive a real-time update on a second device. The results help identify delays that may affect the overall user experience.

The application is tested with multiple donations and images of different sizes, and under slow network conditions, to confirm that screens stay responsive and that images are loaded and cached correctly. Testing also considers how the application behaves when network connectivity is interrupted and restored.

Database queries, image loading and real-time subscriptions are reviewed to identify possible performance bottlenecks. The application should avoid unnecessary repeated requests and should provide suitable loading indicators while data is being retrieved. These tests help evaluate whether the application can handle regular donation activities without making the interface unresponsive.

8.5 USABILITY TESTING

Usability testing evaluates how easy the application is for donors, NGO staff and biogas partners. Since many users are not technical, the screens should be simple, clearly labelled and consistent. The main objective is to ensure that users can understand the available options and complete their tasks without unnecessary difficulty.

During usability testing, users register, log in, create or request a donation, read notifications and view history. The tests confirm that each role sees only the options meant for it and can complete its tasks without help. Particular attention is given to role-specific dashboards, donation forms, expiry information and request status messages.

Feedback from users is collected to improve the layout and wording without redesigning the interface, making the application more intuitive and accessible. Testing also checks whether error messages explain what went wrong and whether empty feeds provide enough information to guide users. The results can be used to improve navigation, reduce confusion and make the donation process easier to follow.

CHAPTER 9
RESULT AND DISCUSSION

The comparison between the existing manual practice and the proposed EcoFeed application shows how the proposed approach improves the handling of surplus food. The comparison is qualitative and is based on the features and workflow of the system rather than on numerical accuracy figures, since EcoFeed is a workflow application and not a prediction model.

In terms of coordination, the existing practice depends on calls and messages to find an NGO. EcoFeed publishes a donation once and shows it immediately to the right audience, and realtime updates and notifications reduce the waiting time. Expired food, which is normally discarded, is routed to biogas partners.

Table 9.1 Existing Practice vs EcoFeed

Capability	Existing	EcoFeed
Structured donation with photo, quantity, expiry	❌	✅
Separate feeds for usable and expired food	❌	✅
Expired food routed to biogas partners	❌	✅
Request, accept/reject and status tracking	❌	✅
Instant notifications and real-time updates	❌	✅
Persistent history across logins and devices	❌	✅
Role-based access enforced in the database	❌	✅

The comparison highlights how EcoFeed organizes the donation process within a single Android application. Instead of relying entirely on manual communication, users can access structured records and follow donation requests through their respective dashboards. Separating edible food from expired or unsuitable food also helps ensure that different categories are directed to the appropriate recipients.

Table 9.2 Role-Based Access Matrix

Action	Food Donor	NGO	Biogas Partner
Create and manage own donations	✅	❌	❌
View non-expired donations	**Own**	✅	❌
View / process expired donations	**Own**	❌	✅
Send request for food	❌	✅	✅
Accept / reject requests	✅	❌	❌
Receive notifications	✅	✅	✅

The role-based design gives EcoFeed a clear advantage in security and accountability. Because the role is stored in public.profiles and the policies are applied in PostgreSQL, a user cannot gain donor rights by changing the interface. Broad policies that let every authenticated user read or modify everything are removed, and the service-role key is never stored in the application.

The development also showed the importance of reliable role handling. In an earlier build, NGO and biogas partner accounts sometimes opened the donor dashboard, and invalid role values caused a validation error. The corrective approach is to use the canonical roles, reject invalid roles instead of defaulting, and clear state when accounts are switched. Each role and account switch must still be verified on real devices, because a successful build alone does not prove correct runtime behaviour.

CHAPTER 10
CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

EcoFeed was developed to reduce food waste by connecting food donors, NGOs and biogas partners through an Android application. Traditional ways of handling surplus food depend on calls and personal contacts, do not separate usable food from expired food and leave no record of what happened to a donation. These limitations can delay food collection and make it difficult to coordinate redistribution and waste processing.

The proposed system addresses these challenges using Kotlin, Jetpack Compose and Supabase. Food donors create donations with photographs, food with a future expiry date is shown to NGOs, expired food is shown to biogas partners, and requests, statuses, notifications and history are saved in the database and updated in real time. This workflow helps users manage donation activities through a centralized platform and improves visibility into the progress of each donation.

Security is handled by role-based access control and PostgreSQL Row Level Security, so that NGOs and biogas partners cannot create donations and users can access only their own profiles and notifications. Testing across the three roles and account switching is essential to confirm this behaviour at runtime. The application also provides a foundation for tracking donation activity and improving communication between users.

In addition, the system can be extended in the future to support more users, locations and waste-processing partners. The consistent use of Supabase for authentication, database, storage and real-time services reduces architectural complexity and removes the need for the older Express and MongoDB backend. Overall, EcoFeed aims to support timely food redistribution, reduce unnecessary food disposal and encourage the productive use of organic waste through biogas and composting.

10.2 FUTURE ENHANCEMENTS

Although EcoFeed provides the main workflow, several improvements can be added in the future to extend its functionality and reach. These enhancements can make the application more convenient for users and improve coordination between donors, NGOs and biogas partners.

One possible enhancement is push notifications through a mobile messaging service, so that users are alerted even when the application is closed. This would shorten response times for perishable food and help users act promptly when a donation request is received or accepted. Notification preferences could also be introduced to let users manage which updates they receive.

Another improvement is location-based matching, in which donations are shown to the nearest NGOs and biogas partners, with map support and pickup routes. This would reduce transport time and cost. The system could also prioritize donations based on distance, expiry time and recipient availability, helping users identify suitable collection opportunities more efficiently.

The system can also include analytics dashboards showing the quantity of food redistributed and sent for biogas, and downloadable reports for donors and partners. These reports could provide useful information about donation trends and the amount of organic waste processed through the platform. Verification of NGOs and biogas partners by an administrator would further increase trust and help maintain reliable participation.

Additionally, an iOS version, multilingual support and an admin panel for monitoring users and donations can be introduced to support wider adoption. Other improvements could include better donation search filters, automated reminders for pending requests and more detailed activity reports. By implementing these enhancements, EcoFeed can become a more complete, efficient and practical solution for reducing food waste and supporting the community.

APPENDIX – A
SOURCE CODE

build.gradle.kts

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.compose)
    alias(libs.plugins.google.services)
    alias(libs.plugins.kotlin.serialization)
}

android {
    namespace = "com.example.ecofeed"
    compileSdk = 36

    defaultConfig {
        applicationId = "com.ecofeed.app"
        minSdk = 26
        targetSdk = 36
        versionCode = 2
        versionName = "1.0.2"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
        )
        }
    }
```

LoginScreen.kt

```
package com.example.ecofeed.ui.auth
import androidx.compose.ui.text.input.VisualTransformation
import androidx.compose.ui.unit.dp
```

```
@Composable
fun LoginScreen(
    viewModel: AuthViewModel,
    onNavigateToSignUp: () -> Unit,
    onLoginSuccess: (String) -> Unit
) {
    val uiState by viewModel.uiState.collectAsState()
    var email by remember { mutableStateOf("") }
    var password by remember { mutableStateOf("") }
    var passwordVisible by remember { mutableStateOf(false) }

    Box(
        modifier = Modifier
            .fillMaxSize()
            .background(MaterialTheme.colorScheme.background)
    ) {
        Column(
            modifier = Modifier
                .fillMaxSize()
                .verticalScroll(rememberScrollState())
                .padding(24.dp),
            verticalArrangement = Arrangement.Top,
            horizontalAlignment = Alignment.Start
        )
                text = "Login to continue reducing food waste",
                style = MaterialTheme.typography.bodyMedium,
                color = MaterialTheme.colorScheme.onSurfaceVariant
            )
```

RoleSelectionScreen.kt

```
package com.example.ecofeed.ui.auth

object AppRoles {
    const val FOOD_DONOR = "FOOD_DONOR"
    const val NGO = "NGO"
    const val BIOGAS_OWNER = "BIOGAS_OWNER"

    fun normalize(role: String?): String = when (role?.uppercase()) {
        "DONOR", "FOOD_DONOR" -> FOOD_DONOR
        "NGO", "CHARITY" -> NGO
        "BIOGAS", "BIOGAS_OWNER", "COMPOST" -> BIOGAS_OWNER
        else -> FOOD_DONOR
    }
}
@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun RoleSelectionScreen(
    viewModel: AuthViewModel,
    onBack: () -> Unit,
    onRegistrationComplete: (String) -> Unit
) {
    var selectedRole by remember { mutableStateOf(AppRoles.FOOD_DONOR) }
    val roles = listOf(
        Triple(AppRoles.FOOD_DONOR, "Food Donor", "Restaurants, Hotels, Caterers,
Supermarkets"),
        Triple(AppRoles.NGO, "NGO / Charity Partner", "Orphanages, Shelters,
Community Kitchens"),
        Triple(AppRoles.BIOGAS_OWNER, "Biogas / Composting Partner", "Organic
Waste Processing Units")
    )
```

DonorDashboardScreen.kt

```
package com.example.ecofeed.ui.donor

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun DonorDashboardScreen(
    onDonateClick: () -> Unit = {},
    onHistoryClick: () -> Unit = {},
    authViewModel: AuthViewModel,
    donationViewModel: DonationViewModel = viewModel()
) {
    val authState by authViewModel.uiState.collectAsState()
    val donationState by donationViewModel.uiState.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("Donor Dashboard", color = Color.White) },
                colors = TopAppBarDefaults.topAppBarColors(containerColor =
PrimaryGreen)
            )
        },
        floatingActionButton = {
            FloatingActionButton(onClick = onDonateClick, containerColor =
PrimaryGreen) {
                Icon(Icons.Default.Add, contentDescription = "Add Donation", tint =
Color.White)
            }
        }
```

NgoDashboardScreen.kt

```
package com.example.ecofeed.ui.ngo

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun NgoDashboardScreen(

    onNavigateToDetail: (String) -> Unit,
    onViewMapClicked: () -> Unit,
    viewModel: NgoViewModel = viewModel()
) {
    val uiState by viewModel.uiState.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("NGO Food Feed", color = Color.White) },
                actions = {
                    IconButton(onClick = onViewMapClicked) {
                        Icon(Icons.Default.Map, contentDescription = "View Map", tint =
Color.White)
                    }
                },
                colors = TopAppBarDefaults.topAppBarColors(containerColor =
PrimaryGreen)
            )
        }
    )
```

```
        Text("Available Surplus Donations", fontWeight = FontWeight.Bold, style =
MaterialTheme.typography.bodySmall)
                }
            }
        }
    }
```

BiogasDashboardScreen.kt

```
package com.example.ecofeed.ui.biogas

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun BiogasDashboardScreen(
    biogasPartnerId: String,
    onNavigateToDetail: (String) -> Unit,
    viewModel: BiogasViewModel = viewModel()
) {
    val uiState by viewModel.uiState.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("Biogas Waste Feed", color = Color.White) },
                colors = TopAppBarDefaults.topAppBarColors(containerColor =
PrimaryGreen)
            )
        }
    ) { innerPadding ->
        Column(
            modifier = Modifier
                .fillMaxSize()
                .padding(innerPadding)
                .padding(16.dp)
```

```
        ) {
            Text("Inedible / Expired Food for Bioconversion", fontWeight =
FontWeight.Bold, style = MaterialTheme.typography.titleMedium)
            Spacer(modifier = Modifier.height(12.dp))

            LazyColumn(verticalArrangement = Arrangement.spacedBy(12.dp)) {
                items(uiState.biogasListings) { listing ->
}
```

ActivityHistoryScreen.kt

```
package com.example.ecofeed.ui.history


@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun ActivityHistoryScreen(
    userRole: String,
    onBack: () -> Unit,
    itemsList: List<HistoryItemDto>
) {
    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("Activity History", color = Color.White) },
                navigationIcon = {
                    IconButton(onClick = onBack) {
                        Icon(Icons.AutoMirrored.Filled.ArrowBack, contentDescription =
"Back", tint = Color.White)
                    }
                },
                colors = TopAppBarDefaults.topAppBarColors(containerColor =
PrimaryGreen)
            )
        }
```

NotificationSyncManager.kt

```
package com.example.ecofeed.service

object NotificationSyncManager {
    private const val TAG = "EcoFeedNotifSync"
    private val scope = CoroutineScope(Dispatchers.IO + SupervisorJob())
    private val repository = SupabaseRepository()
    private var syncJob: Job? = null
    private val _unreadCount = MutableStateFlow(0)
    val unreadCount: StateFlow<Int> = _unreadCount.asStateFlow()
    fun start(context: Context) {
        val appContext = context.applicationContext
        if (syncJob?.isActive == true) return

        syncJob = scope.launch {
            val tracker = NotificationTracker(appContext)
            val prefs = UserPreferences(appContext)

            while (isActive) {
                try {
                    doSync(appContext, tracker, prefs)
                } catch (e: Exception) {
                    Log.e(TAG, "Error in notification sync loop: ${e.message}")
                }
                delay(3000)
            }
        }
```

supabase_schema.sq

```
CREATE TABLE IF NOT EXISTS public.profiles (
    id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    name TEXT NOT NULL DEFAULT '',
    full_name TEXT DEFAULT '',
    email TEXT NOT NULL DEFAULT '',
    phone TEXT DEFAULT '',
    role TEXT DEFAULT 'NGO Member',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text,
NOW()) NOT NULL,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text,
NOW()) NOT NULL
);
ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
DROP POLICY IF EXISTS "Public profiles are viewable by everyone" ON
public.profiles;
CREATE POLICY "Public profiles are viewable by everyone" ON public.profiles
    FOR SELECT USING (true);
CREATE TABLE IF NOT EXISTS public.donations (
    id TEXT PRIMARY KEY,
    donor_id TEXT,
    title TEXT NOT NULL,
    quantity TEXT NOT NULL,
    quantity_kg DOUBLE PRECISION DEFAULT 0.0,
    category TEXT DEFAULT 'General',
    expiry_date TEXT NOT NULL,
    status TEXT NOT NULL DEFAULT 'AVAILABLE',
    accepted_by TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text,
NOW()) NOT NULL
);
```

APPENDIX – B

SCREENSHOTS

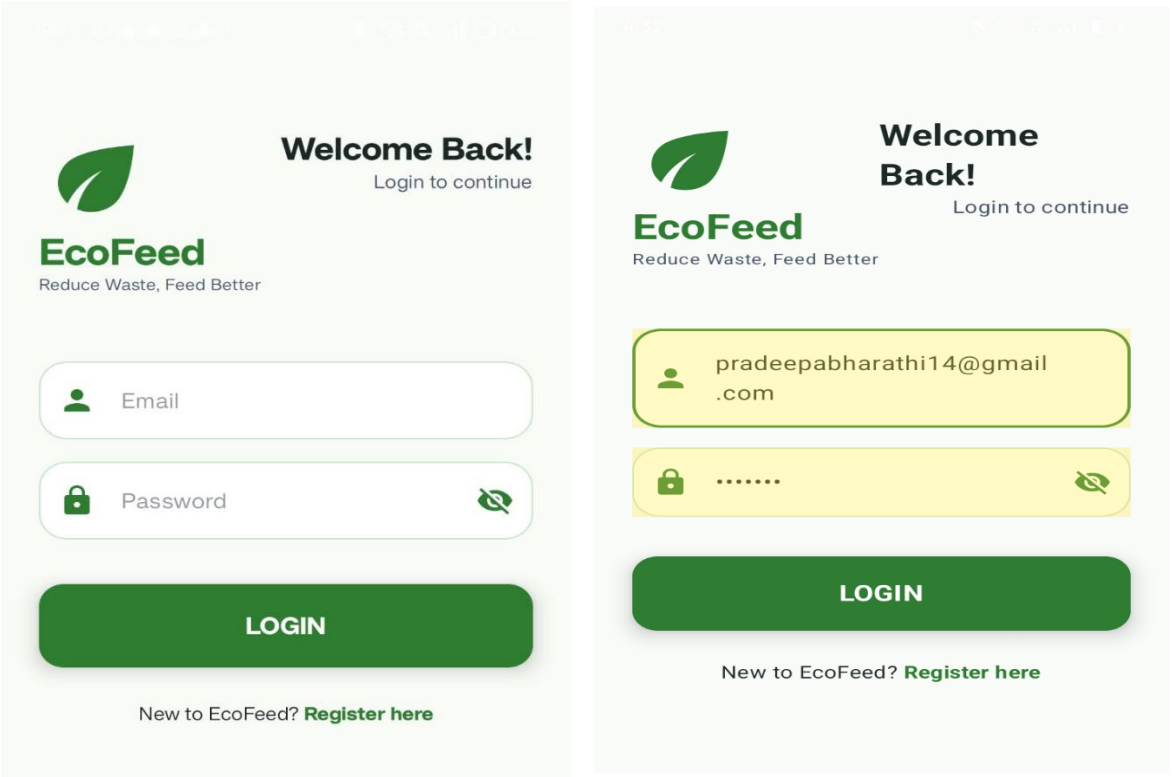


Figure B.1 Login and Registration Screen

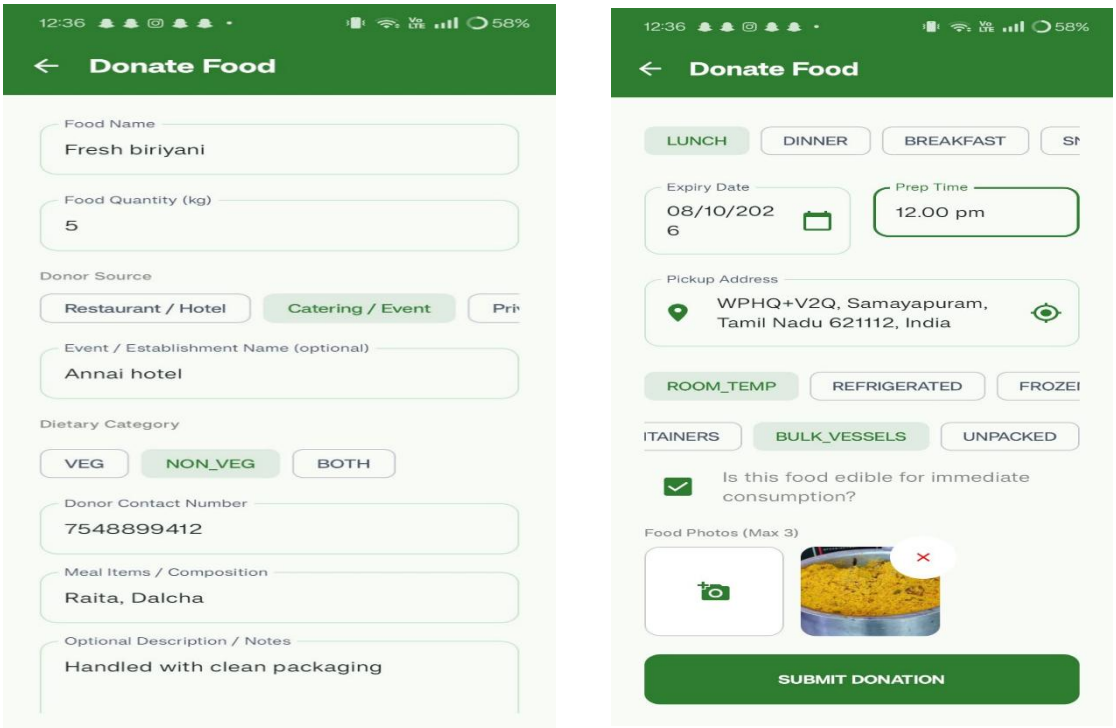


Figure B.2 Food Donor Dashboard

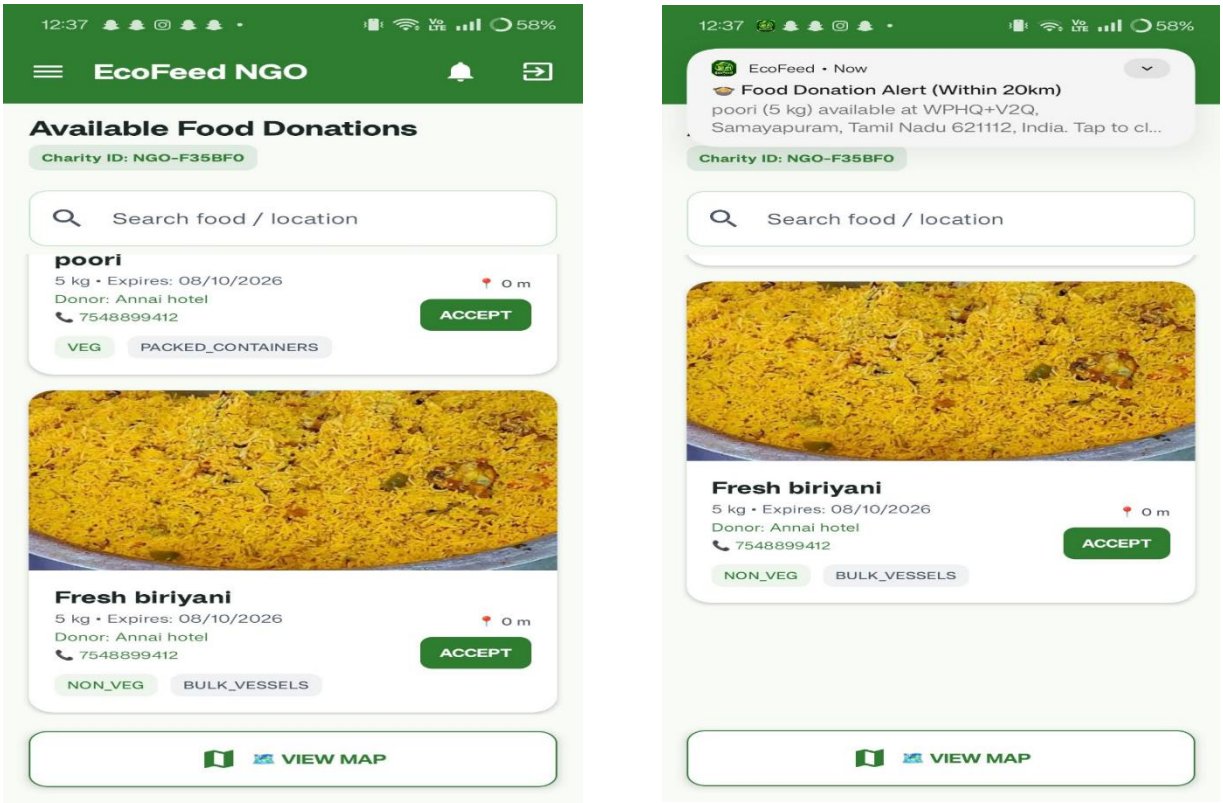


Figure B.3 NGO Dashboard

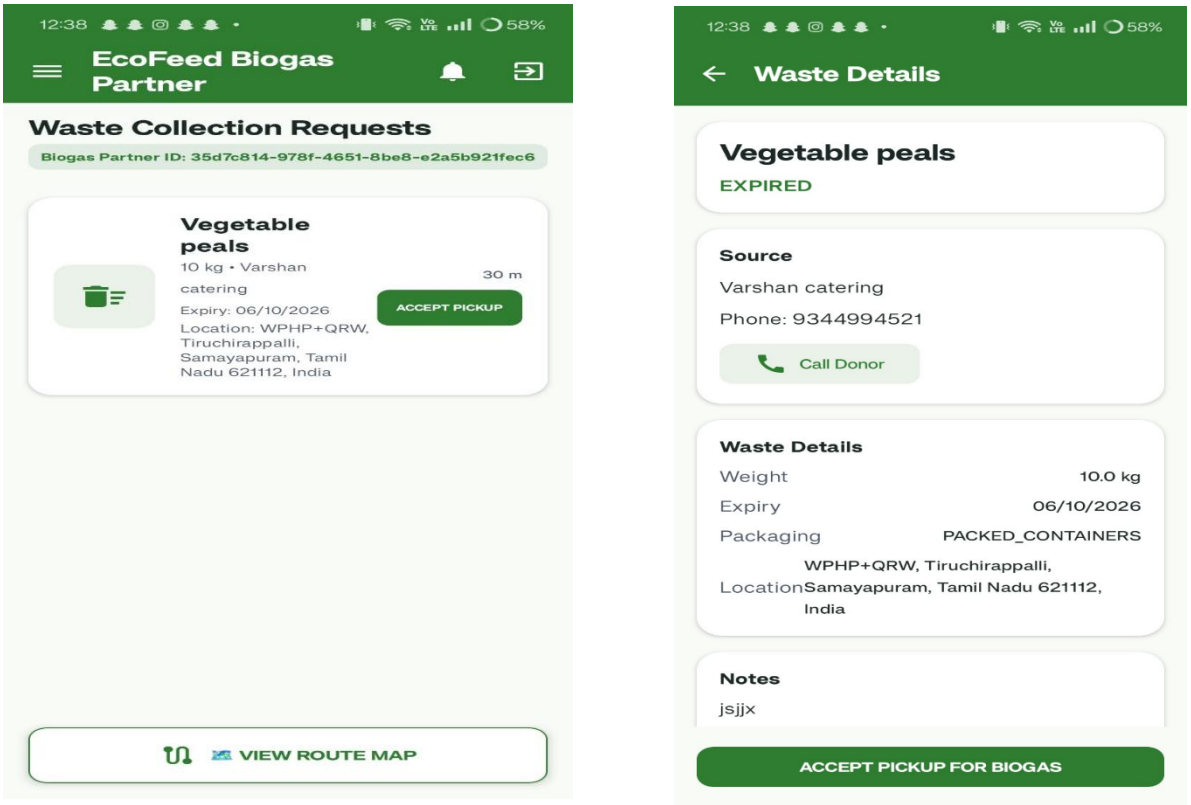


Figure B.4 Biogas Partner Dashboard

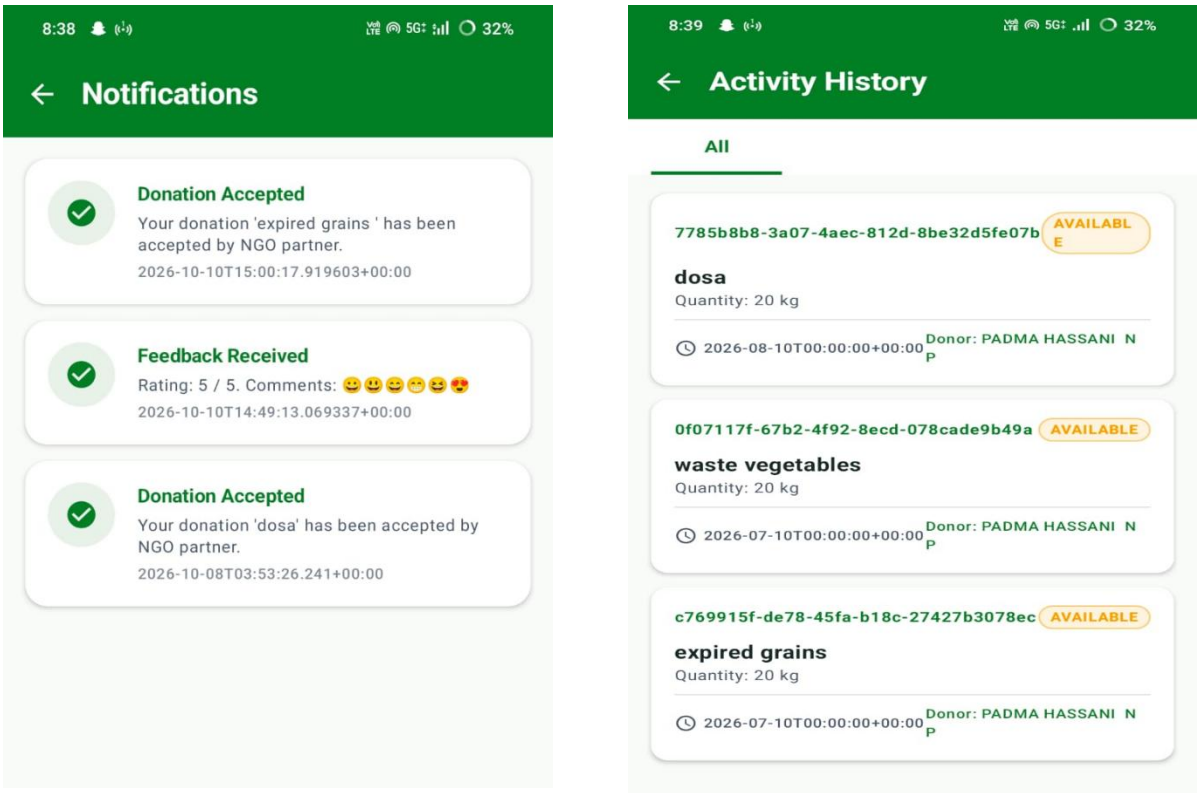


Figure B.5 Notifications and Activity History

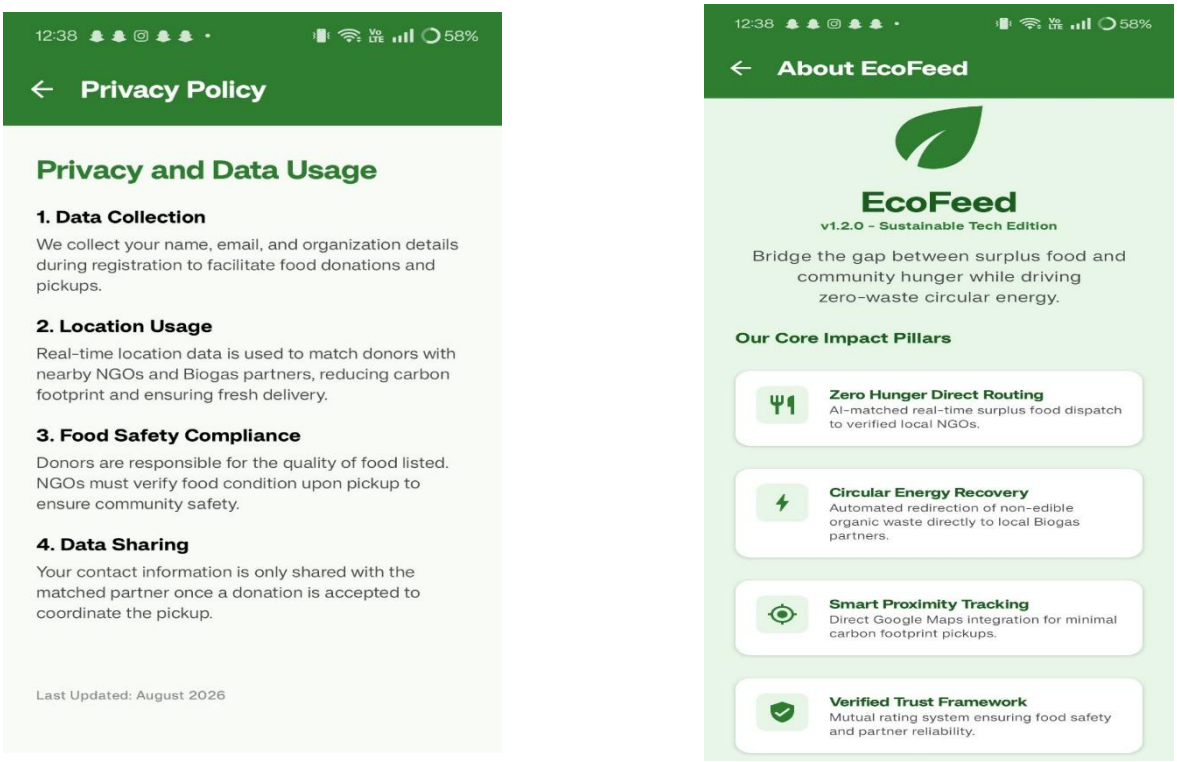


Figure B.6 Privacy Policy and About EcoFeed

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